

Digital Filter Theory & Mathematical Formulations

CONTINUOUS PROTOTYPES, BILINEAR MAPPING, AND SECOND-ORDER SECTIONS (SOS)

1. Theoretical Foundations & Overview

Linear time-invariant (LTI) digital filtering is a foundational subject in discrete-time signal processing. Classical infinite impulse response (IIR) filter synthesis translates well-understood continuous-time prototype transfer functions into numerically stable discrete-time difference equations.

In practice, the modern synthesis pipeline comprises three systematic stages:

1. **Analog Prototype Approximation:** Deriving an all-pole or pole-zero continuous transfer function $H_a(s)$ satisfying magnitude and phase specifications in the normalized continuous frequency domain.
2. **Continuous-to-Discrete Mapping:** Applying the bilinear transformation with non-linear tangent frequency pre-warping to preserve critical passband and stopband boundaries.
3. **Second-Order Section (SOS) Decomposition:** Factoring high-order polynomials into cascaded biquads to eliminate coefficient quantization sensitivity in real-time execution.

2. Continuous-Time Prototype Filter Families

2.1 Butterworth Approximation (Maximally Flat)

The Butterworth filter is characterized by a maximally flat magnitude response in the passband. The squared-magnitude frequency response is given by:

$$|H(j\Omega)|^2 = \frac{1}{1 + \left(\frac{\Omega}{\Omega_c}\right)^{2N}} \quad (1)$$

where N denotes the filter order and Ω_c is the -3.01 dB cutoff frequency. The poles of $H(s)H(-s)$ are distributed symmetrically along a circle of radius Ω_c in the complex s -plane. Causality and stability require selecting the N left-half plane poles:

$$s_k = \Omega_c \exp\left(j\frac{\pi(2k + N - 1)}{2N}\right), \quad k = 1, 2, \dots, N \quad (2)$$

2.2 Chebyshev Approximations (Equiripple Designs)

Chebyshev approximations minimize the peak absolute error across the specified band by utilizing Chebyshev polynomials of the first kind, $T_N(x)$, generated via the recurrence relation $T_0(x) = 1$, $T_1(x) = x$, and $T_{n+1}(x) = 2xT_n(x) - T_{n-1}(x)$.

Type I (Passband Equiripple):

$$|H(j\Omega)|^2 = \frac{1}{1 + \epsilon^2 T_N^2\left(\frac{\Omega}{\Omega_p}\right)} \quad (3)$$

where $\epsilon = \sqrt{10^{R_p/10} - 1}$ parametrizes passband ripple R_p (dB). The poles reside on an ellipse in the complex s -plane.

Type II (Inverse Chebyshev, Stopband Equiripple):

$$|H(j\Omega)|^2 = \frac{1}{1 + [\delta^2 T_N^2\left(\frac{\Omega_s}{\Omega}\right)]^{-1}} \quad (4)$$

The passband is maximally flat, while the stopband exhibits equiripple behavior with transmission zeros placed directly on the imaginary axis $s = \pm j\Omega_{z,k}$.

2.3 Elliptic (Cauer) Rational Filters

Elliptic filters achieve optimal transition steepness for a prescribed order N by permitting equiripple behavior in both passband and stopband:

$$|H(j\Omega)|^2 = \frac{1}{1 + \epsilon^2 R_N^2\left(\xi, \frac{\Omega}{\Omega_p}\right)} \quad (5)$$

where $R_N(\xi, x)$ is the Chebyshev rational function with selectivity parameter $\xi = \Omega_s/\Omega_p > 1$.

2.4 Bessel-Thomson Linear Phase Filters

Bessel filters approximate constant group delay $\tau_g(\Omega) = -d\theta/d\Omega \approx \tau_0$ using reverse Bessel polynomials $B_N(s)$:

$$H(s) = \frac{B_N(0)}{B_N(s/\Omega_0)}, \quad B_N(s) = \sum_{k=0}^N \frac{(2N-k)!}{2^{N-k} k! (N-k)!} s^k \quad (6)$$

Because the phase response is linear across the passband, Bessel filters eliminate transient ringing in step and pulse responses.

Filter Prototype	Passband	Stopband	Transition Width	Group Delay
Butterworth	Maximally Flat	Monotonic	Moderate	Moderately Linear
Chebyshev Type I	Equiripple	Monotonic	Sharp	Non-linear near edge
Chebyshev Type II	Maximally Flat	Equiripple	Sharp	Moderate Distortion
Elliptic (Cauer)	Equiripple	Equiripple	Optimal (Sharpest)	Highly Non-linear
Bessel-Thomson	Monotonic	Monotonic	Gradual	Maximally Flat

Table 1: Comparison of canonical continuous-time prototype approximations.

3. Bilinear Transformation & Frequency Pre-Warping

The bilinear transform maps the entire left-half continuous s -plane into the interior of the discrete unit circle $|z| < 1$ via trapezoidal integration:

$$s = \frac{2}{T_s} \frac{1 - z^{-1}}{1 + z^{-1}} \iff z = \frac{1 + \frac{T_s}{2}s}{1 - \frac{T_s}{2}s} \quad (7)$$

where $T_s = 1/f_s$ is the sampling interval. Evaluating along $s = j\Omega$ and $z = e^{j\omega}$ yields the non-linear frequency warping relation:

$$\Omega = \frac{2}{T_s} \tan\left(\frac{\omega}{2}\right) \iff \omega = 2 \arctan\left(\frac{\Omega T_s}{2}\right) \quad (8)$$

To ensure synthesized discrete cutoff frequencies ω_0 match the engineering targets, continuous prototype frequencies must be pre-warped prior to polynomial transformation:

$$\Omega_{\text{pre}} = 2f_s \tan\left(\frac{\pi f_{\text{target}}}{f_s}\right) \quad (9)$$

4. Second-Order Sections (SOS) Cascading

High-order rational transfer functions $H(z) = B(z)/A(z)$ exhibit extreme coefficient sensitivity, where slight numerical rounding causes catastrophic pole displacement outside the unit circle. Therefore, $H(z)$ is decomposed into a cascade of $K = \lceil N/2 \rceil$ second-order sections:

$$H(z) = g \cdot \prod_{k=1}^K \frac{b_{0,k} + b_{1,k}z^{-1} + b_{2,k}z^{-2}}{1 + a_{1,k}z^{-1} + a_{2,k}z^{-2}} \quad (10)$$

4.1 Direct Form II Transposed Difference Equations

Each biquad section is implemented using the Direct Form II Transposed topology:

$$y_k[n] = b_{0,k}x_k[n] + w_{1,k}[n-1] \quad (11)$$

$$w_{1,k}[n] = b_{1,k}x_k[n] - a_{1,k}y_k[n] + w_{2,k}[n-1] \quad (12)$$

$$w_{2,k}[n] = b_{2,k}x_k[n] - a_{2,k}y_k[n] \quad (13)$$

This topology requires only two delay state registers (w_1, w_2) per biquad and minimizes internal accumulator overflow in real-time execution.

5. Canonical References

- [1] A. V. Oppenheim and R. W. Schaffer, *Discrete-Time Signal Processing*, 3rd ed., Prentice Hall, 2009.
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- [3] S. K. Mitra, *Digital Signal Processing: A Computer-Based Approach*, 4th ed., McGraw-Hill, 2011.
- [4] U. Zölzer, *DAFX: Digital Audio Effects*, 2nd ed., John Wiley & Sons, 2011.